



TRƯỜNG ĐẠI HỌC FPT

ASSIGNMENT

LEAGUE FOOTBALL TOURNAMENTMANAGER - DBI202 -

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Prepared : Group 7

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We would like to express our sincere thanks to **Mr. Phan Ngoc Hoang**, our lecturer of **Database Systems**, for assigning this project to us and for his guidance and support throughout the course. Although he was not directly involved in our project, his lectures and the sample assignment form provided us with useful knowledge and structure to complete this report successfully.

This project offered us an excellent opportunity to apply the database concepts and SQL implementation we learned in class to a real-world context, using the management of football tournaments as the main scenario. By choosing a topic related to football – one of the most popular sports worldwide – we had the opportunity to design a database that reflects practical tournament operations such as team registration, match scheduling, referee assignments, and result tracking.

Through this project, we were able to gain valuable experience, strengthen our understanding of database modeling, and improve our analytical, teamwork, and problem-solving skills.

We would also like to sincerely thank all of our group members for their teamwork, responsibility, and dedication during the development of this report entitled “**Champions League Football Tournament Manager.**” Each member contributed effort and ideas that helped complete the assignment effectively and on time

I) INTRODUCE THE PROBLEM.....	3
1) Describe the problem.....	3
2) Management objectives.....	5
II) ENTITY - RELATIONSHIP - ER.....	6
1) Definition entity – attribute.....	6
2) Set-up entity – relationship.....	6
LINK:	
https://drive.google.com/file/d/1geDMOfREXR-FdY8qL-PE4_8N0EuwWwfn/view?usp=sharing	9
III) DATA DICTIONARY.....	10
1) Database and Table.....	10
A. CREATE DATABASE PROJECTDBI202.....	10
B. CREATE TABLE ORGANIZER.....	10
C. CREATE TABLE TEAMS.....	11
D. CREATE TABLE PLAYERS.....	13
E. CREATE TABLE STADIUMS.....	14
F. CREATE TABLE STAFF.....	16
G. CREATE TABLE REFEREES.....	17
H. CREATE TABLE MATCHES.....	18
I. CREATE TABLE GOALS.....	20
J. CREATE TABLE CARDS.....	21
K. CREATE TABLE CLUB-MATCHES.....	22
IV. ENTITY RELATIONSHIP DIAGRAM (ERD).....	23
A. ORGANIZER.....	23
B. CLUBS.....	24
C. PLAYERS.....	26
D. STADIUMS.....	27
E. STAFF.....	28
F. REFEREES.....	29
G. MATCHES.....	30
H. GOALS.....	31
I. CARDS.....	31
J. CLUB_MATCHES.....	32
K. FULL DIAGRAM.....	33
V. SQL COMMAND.....	34
A. query using order by.....	34
B. query using inner join.....	34
C. query using aggregate functions.....	35

D. query using the group by and having clauses.....	36
E. query that uses a sub-query as a relation.....	37
F. query that uses partial matching in the where clause.....	38
G. query that uses a self – join.....	39
H. store procedure.....	39
I. trigger.....	40

I) INTRODUCE THE PROBLEM

1) Describe the problem

The UEFA Champions League is one of the most prestigious football tournaments in the world, including many famous football clubs across Europe. However, there is still no centralized database system to manage all the essential information such as clubs, players, referees, stadiums, matches, goals, and cards.

After researching and analyzing the tournament structure, the current situation can be described as follows:

- Each **organizer** is responsible for managing the tournament and the participation of all clubs.
- Each **club** has only one coach (manager) and many registered players. Each player belongs to exactly one club.
- Information about the club includes: club name, country, coach name, coach phone, coach email, and organizer.
- Information about the player includes: full name, nationality, position, shirt number, phone, email, and the club they belong to.
- Each **stadium** has information such as name, city, capacity, phone, and email.
- Each stadium has several **staff members**, including ground managers, security, medical, and technicians.

- The **referees** table stores information of referees, including full name, nationality, phone, and email.
- Each **match** takes place in one stadium, managed by one referee, and involves two different clubs (home and away).
- Each match has information such as match date, stage (Quarter-final, Semi-final, Final, etc.), referee, stadium, home team, away team, and score.
- Each **goal** is linked to a specific player and match, with details such as the minute and goal type (Normal, Header, Penalty, etc.).
- Each **card** is also linked to a player and match, including card type (Yellow or Red) and the minute the card was given.
- A separate table is used to record which clubs participated in which matches, ensuring clear tracking of tournament participation.

Request:

- Allow registration of organizers, clubs (with their coaches), players, referees, stadiums, and stadium staff.
- Allow creation and storage of match schedules in the **Matches** table, including match date, stage, stadium, referee, home club, away club, and score.
- Allow input of match results by recording goals (**Goals** table) and cards (**Cards** table) after each match.
- Allow users to view all players belonging to each club and all clubs participating in each match.

- Allow users to display all matches hosted by each stadium and all matches managed by each referee.
- Allow users to view stadium staff information and the stadiums they are responsible for.
- Allow users to check detailed match events, such as goals and cards, for each match or each player.
- Allow users to list matches by tournament stage (e.g., Quarter-Final, Semi-Final, Final).

Therefore, a centralized Champions League Tournament Management System is required to ensure accurate data tracking, efficient scheduling, and transparent competition results.

2) Management objectives

- Manage information of organizers, clubs, coaches, and players.
- Manage stadiums and their staff.
- Manage referees and their assigned matches.
- Manage tournament matches, stages, and results.
- Manage goal and card statistics for each match and player.
- Manage the relationship between clubs and matches they participate in.

II) ENTITY - RELATIONSHIP - ER

1) Definition entity – attribute

Base on the problem description and management objectives, we can present several entities and attributes of the entity as follow:

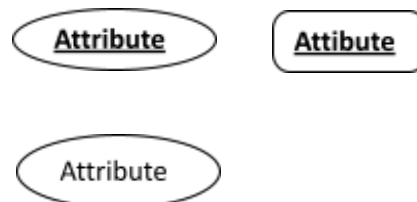
- Organizer: **OrganizerID**, OrganizerName, Contact, Phone, Email.

- Clubs: **ClubID**, ClubName, Country, CoachName, CoachPhone, CoachEmail, **OrganizerID**.
- Players: **PlayerID**, FullName, Nationality, Position, ShirtNumber, Phone, Email, **ClubID**.
- Stadiums: **StadiumID**, StadiumName, City, Capacity, Phone, Email.
- Staff: **StaffID**, FullName, Role, Phone, Email, **StadiumID**.
- Referees: **RefereeID**, FullName, Country, Phone, Email.
- Matches: **MatchID**, MatchDate, Stage, **StadiumID**, **RefereeID**, **HomeClubID**, **AwayClubID**, Score.
- Goals: **GoalID**, **MatchID**, **PlayerID**, Minute, GoalType.
- Cards: **CardID**, **MatchID**, **PlayerID**, CardType, Minute.
- Club_Matches: **ClubID**, **MatchID**.

2) Set-up entity – relationship

* Some symbols used in the model

- Key / identifier attribute



- Attribute description / description

Attribute

ENTITY

- Entity

WEAK ENTITY

- Weak entity

Relationship

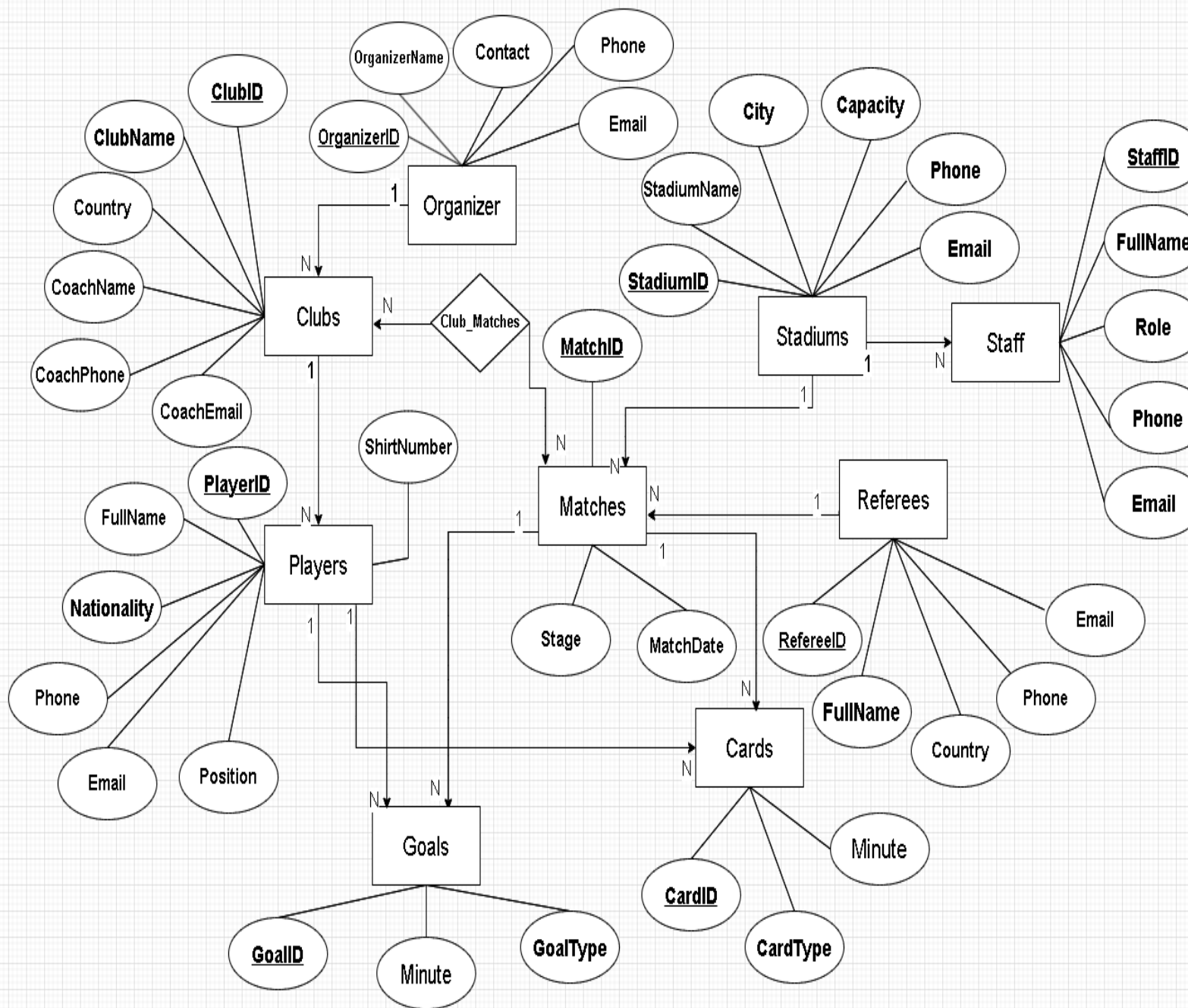
- Relationship

- Connectivity (force) = 1



- Connectivity = N





LINK:

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III) DATA DICTIONARY

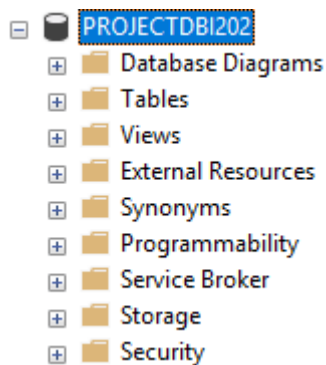
Just for example on some tables (other tables are similar, you have to define all the tables in your database). Note: to run the query you have to define the table 1 first then go to the side tables much

1) Database and Table

A. CREATE DATABASE PROJECTDBI202

--create database

CREATE DATABASE PROJECTDBI202



B. CREATE TABLE ORGANIZER

Column Name	Data Type	Default	Key/ Index/ Constraint
OrganizerID	Int		IDENTITY(1,1), PRIMARY KEY
OrganizerName	NVARCHAR(100)		NOT NULL
Contact	NVARCHAR(100)		

Column Name	Data Type	Default	Key/ Index/ Constraint
Phone	NVARCHAR(25)		NOT NULL
Email	NVARCHAR(100)		NOT NULL

Code:

-- 1. ORGANIZER

```
CREATE TABLE Organizer (
    OrganizerID INT IDENTITY(1,1),
    OrganizerName NVARCHAR(100) NOT NULL,
    Contact NVARCHAR(100),
    Phone NVARCHAR(25) not null,
    Email NVARCHAR(100) not null,
    PRIMARY KEY (OrganizerID),
)
```

Example:

OrganizerID	OrganizerName	Contact	Phone	Email
1	UEFA	Champions League 2024-2025	+41 22 123 4567	info@uefa.com

C. CREATE TABLE TEAMS

Column Name	Data Type	Default	Key/ Index/ Constraint
ClubID	INT		IDENTITY(1,1), PRIMARY KEY
ClubName	NVARCHAR(100)		NOT NULL

Column Name	Data Type	Default	Key/ Index/ Constraint
Country	NVARCHAR (50)		
CoachName	NVARCHAR (100)		
CoachPhone	NVARCHAR (25)		
CoachEmail	NVARCHAR (100)		
OrganizerID	INT		FOREIGN KEY

Code:

-- 2. TEAMS

```
CREATE TABLE Clubs (
    ClubID INT IDENTITY(1,1),
    ClubName NVARCHAR(100) NOT NULL,
    Country NVARCHAR(50),
    CoachName NVARCHAR(100),
    CoachPhone NVARCHAR(25),
    CoachEmail NVARCHAR(100),
    OrganizerID INT,
    PRIMARY KEY (ClubID),
    FOREIGN KEY (OrganizerID) REFERENCES Organizer(OrganizerID)
)
```

Example:

ClubID	ClubName	Country	CoachName	CoachPhone	CoachEmail	OrganizerID
1	Real Madrid	Spain	Carlo Ancelotti	+34 600 123 456	ancelotti@realmadrid.com	1
2	Manchester City	England	Pep Guardiola	+44 700 234 567	pep@mcfc.co.uk	1
3	Bayern Munich	Germany	Thomas Tuchel	+49 171 987 654	tuchel@fcbayern.de	1
4	Paris Saint-Germain	France	Luis Enrique	+33 612 345 678	luis.enrique@psg.fr	1
5	Inter Milan	Italy	Simone Inzaghi	+39 340 222 333	inzaghi@inter.it	1
6	Barcelona	Spain	Xavi Hernandez	+34 611 999 888	xavi@fcbarcelona.com	1
7	Arsenal	England	Mikel Arteta	+44 701 555 666	arteta@arsenal.co.uk	1
8	Liverpool	England	Jürgen Klopp	+44 700 777 888	klopp@liverpoolfc.com	1

D. CREATE TABLE PLAYERS

Column Name	Data Type	Default	Check	Key/ Index/ Constraint
PlayerID	INT			IDENTITY(1,1), Primary key
FullName	NVARCHAR (100)			NOT NULL
Nationality	NVARCHAR (50)			
Position	NVARCHAR (50)			
ShirtNumber	INT			
Phone	NVARCHAR (25)			
Email	NVARCHAR (100)			
ClubID	INT			FOREIGN KEY

Code:

-- 3. PLAYERS

```
CREATE TABLE Players (  
    PlayerID INT IDENTITY(1,1),  
    FullName NVARCHAR(100) NOT NULL,  
    Nationality NVARCHAR(50),  
    Position NVARCHAR(50),  
    ShirtNumber INT,  
    Phone NVARCHAR(25),  
    Email NVARCHAR(100),  
    ClubID INT,  
    PRIMARY KEY(PlayerID),  
    FOREIGN KEY (ClubID) REFERENCES Clubs(ClubID)  
)
```

Example:

PlayerID	FullName	Nationality	Position	ShirtNumber	Phone	Email	ClubID
1	Thibaut Courtois	Belgium	Goalkeeper	1	+34 611 100 001	courtois@realmadrid.com	1
2	Dani Carvajal	Spain	Defender	2	+34 611 100 002	carvajal@realmadrid.com	1
3	Antonio Rüdiger	Germany	Defender	4	+34 611 100 003	rudiger@realmadrid.com	1
4	David Alaba	Austria	Defender	3	+34 611 100 004	alaba@realmadrid.com	1
5	Ferland Mendy	France	Defender	23	+34 611 100 005	mendy@realmadrid.com	1
6	Jude Bellingham	England	Midfielder	5	+34 611 100 006	bellingham@realmadrid.com	1
7	Federico Valverde	Uruguay	Midfielder	15	+34 611 100 007	valverde@realmadrid.com	1
8	Luka Modric	Croatia	Midfielder	10	+34 611 100 008	modric@realmadrid.com	1

E. CREATE TABLE STADIUMS

Column Name	Data Type	Default	Key/ Index/ Constraint
StadiumID	INT		IDENTITY(1,1), PRIMARY KEY
StadiumName	NVARCHAR(100)		NOT NULL
City	NVARCHAR(100)		

Column Name	Data Type	Default	Key/ Index/ Constraint
Capacity	INT		
Phone	NVARCHAR(25)		
Email	NVARCHAR(100)		

Code:

-- 4. STADIUMS

```
CREATE TABLE Stadiums (
    StadiumID INT IDENTITY(1,1),
    StadiumName NVARCHAR(100) NOT NULL,
    City NVARCHAR(100),
    Capacity INT,
    Phone NVARCHAR(25),
    Email NVARCHAR(100),
    PRIMARY KEY(StadiumID)
)
```

Example:

StadiumID	StadiumName	City	Capacity	Phone	Email
1	Santiago Bernabeu	Madrid	81044	+34 600 700 800	bernabeu@realmadrid.com
2	Etihad Stadium	Manchester	53400	+44 700 800 900	etihad@mcf.co.uk
3	Allianz Arena	Munich	75000	+49 800 900 100	allianz@fcbayern.de
4	Parc des Princes	Paris	47929	+33 900 100 200	parc@psg.fr

F. CREATE TABLE STAFF

Column Name	Data Type	Default	Check	Key/ Index/ Constraint
StaffID	INT			IDENTITY(1,1), PRIMARY KEY
FullName	NVARCHAR(100)			NOT NULL
Role	NVARCHAR(50)			
Phone	NVARCHAR(25)			
Email	NVARCHAR(100)			
StadiumID	INT			FOREIGN KEY

Code:

-- 5. STAFF (thuộc Stadiums)

```
CREATE TABLE Staff (
    StaffID INT IDENTITY(1,1),
    FullName NVARCHAR(100) NOT NULL,
    Role NVARCHAR(50),
    Phone NVARCHAR(25),
    Email NVARCHAR(100),
    StadiumID INT,
    PRIMARY KEY(StaffID),
    FOREIGN KEY (StadiumID) REFERENCES Stadiums(StadiumID)
)
```

Example:

StaffID	FullName	Role	Phone	Email	StadiumID
1	Miguel Santos	Ground Manager	+34 700 111 001	miguel@realmadrid.com	1
2	Carlos Perez	Security	+34 700 111 002	carlos@realmadrid.com	1
3	Ana Gomez	Medical	+34 700 111 003	ana@realmadrid.com	1
4	Luis Torres	Technician	+34 700 111 004	luis@realmadrid.com	1
5	John Taylor	Ground Manager	+44 800 222 001	john@mcfc.co.uk	2
6	Emma Collins	Security	+44 800 222 002	emma@mcfc.co.uk	2

G. CREATE TABLE REFEREES

Column Name	Data Type	Default	Key/ Index/ Constraint
RefereeID	INT		IDENTITY(1,1), PRIMARY KEY
FullName	NVARCHAR(100)		NOT NULL
Country	NVARCHAR(50)		
Phone	NVARCHAR(25)		
Email	NVARCHAR(100)		

Code:

-- 6. REFEREES

```
CREATE TABLE Referees (
    RefereeID INT IDENTITY(1,1),
    FullName NVARCHAR(100) NOT NULL,
    Country NVARCHAR(50),
    Phone NVARCHAR(25),
    Email NVARCHAR(100),
    PRIMARY KEY(RefereeID)
)
```

Example:

RefereeID	FullName	Country	Phone	Email
1	Daniele Orsato	Italy	+39 340 111 222	orsato@uefa.com
2	Clément Turpin	France	+33 612 111 222	turpin@uefa.com
3	Anthony Taylor	England	+44 701 111 222	taylor@uefa.com
4	Danny Makkelie	Netherlands	+31 612 333 444	makkelie@uefa.com
5	Szymon Marciniak	Poland	+48 600 555 666	marciniak@uefa.com

H. CREATE TABLE MATCHES

Column Name	Data Type	Default	Check	Key/ Index/ Constraint
MatchID	INT			IDENTITY(1,1), PRIMARY KEY
MatchDate	DATE			NOT NULL
Stage	NVARCHAR(50)			
StadiumID	INT			FOREIGN KEY
RefereeID	INT			FOREIGN KEY
HomeClubID	INT			FOREIGN KEY
AwayClubID	INT			FOREIGN KEY

Column Name	Data Type	Default	Key/ Index/ Constraint
Score	NVARCHAR(20)		

Code:

-- 7. MATCHES

```
CREATE TABLE Matches (
    MatchID INT IDENTITY(1,1),
    MatchDate DATE NOT NULL,
    Stage NVARCHAR(50),
    StadiumID INT,
    RefereeID INT,
    HomeClubID INT,
    AwayClubID INT,
    Score NVARCHAR(20),
    PRIMARY KEY(MatchID),
    FOREIGN KEY (StadiumID) REFERENCES Stadiums(StadiumID),
    FOREIGN KEY (RefereeID) REFERENCES Referees(RefereeID),
    FOREIGN KEY (HomeClubID) REFERENCES Clubs(ClubID),
    FOREIGN KEY (AwayClubID) REFERENCES Clubs(ClubID)
)
```

Example:

MatchID	MatchDate	Stage	StadiumID	RefereeID	HomeClubID	AwayClubID	Score
1	2025-04-08	Quarter-Final	1	1	1	8	3-1
2	2025-04-09	Quarter-Final	2	2	2	5	2-0
3	2025-04-10	Quarter-Final	3	3	3	6	1-1
4	2025-04-11	Quarter-Final	4	4	4	7	2-2
5	2025-04-29	Semi-Final	1	5	1	2	2-1
6	2025-04-30	Semi-Final	3	6	3	4	1-2
7	2025-05-28	Final	3	7	1	4	2-0

I. CREATE TABLE GOALS

Column Name	Data Type	Default	Check	Key/ Index/ Constraint
GoalID	INT			IDENTITY(1,1), PRIMARY KEY
MatchID	INT			FOREIGN KEY
PlayerID	INT			FOREIGN KEY
Minute	INT			
GoalType	NVARCHAR(20)			

Code:

-- 8. GOALS (phụ thuộc MATCH và PLAYER)

```
CREATE TABLE Goals (  
    GoalID INT IDENTITY(1,1),  
    MatchID INT,  
    PlayerID INT,  
    Minute INT,  
    GoalType NVARCHAR(20),  
    PRIMARY KEY(GoalID),  
    FOREIGN KEY (MatchID) REFERENCES Matches(MatchID),  
    FOREIGN KEY (PlayerID) REFERENCES Players(PlayerID)  
)
```

Example:

GoalID	MatchID	PlayerID	Minute	GoalType
1	1	9	17	Normal
2	1	10	45	Penalty
3	1	3	78	Header
4	5	6	34	Normal

J. CREATE TABLE CARDS

Column Name	Data Type	Default	Check	Key/ Index/ Constraint
CardID	INT			IDENTITY(1,1), PRIMARY KEY
MatchID	Int			FOREIGN KEY
PlayerID	INT			FOREIGN KEY
CardType	NVARCHAR(10)			
Minute	INT			

Code:

-- 9. CARDS (phụ thuộc MATCH và PLAYER)

```
CREATE TABLE Cards (
    CardID INT IDENTITY(1,1),
    MatchID INT,
    PlayerID INT,
    CardType NVARCHAR(10), -- Yellow, Red
    Minute INT,
    PRIMARY KEY(CardID),
    FOREIGN KEY (MatchID) REFERENCES Matches(MatchID),
    FOREIGN KEY (PlayerID) REFERENCES Players(PlayerID)
)
```

Example:

CardID	MatchID	PlayerID	CardType	Minute
1	1	2	Yellow	55
2	1	8	Red	70
3	5	6	Yellow	42
4	7	9	Yellow	50

K. CREATE TABLE CLUB-MATCHES

Column Name	Data Type	Default	Check	Key/ Index/ Constraint
ClubID	INT			NOT NULL, PK,FK
MatchID	Int			NOT NULL, PK ,FK

Code:

-- 10. (Tùy chọn) Bảng trung gian nhiều đội tham gia 1 trận

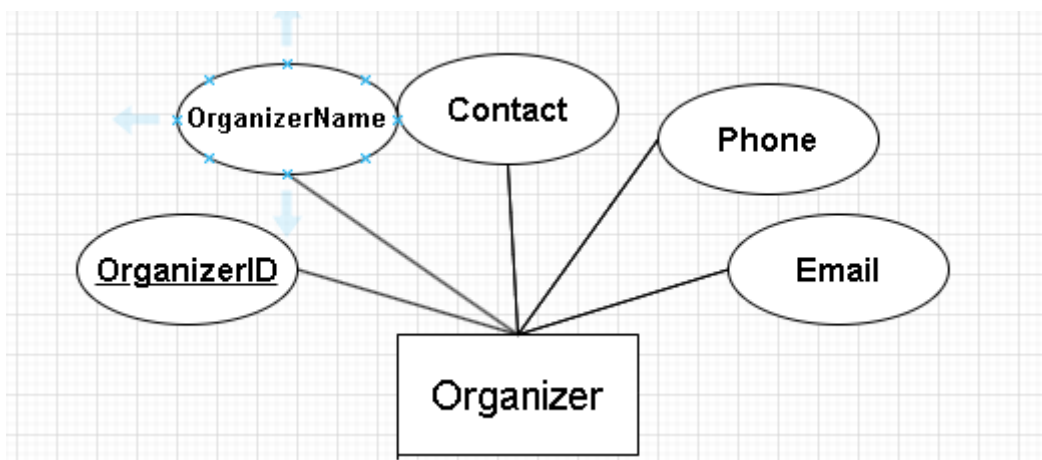
```
CREATE TABLE Club_Matches (
    ClubID INT NOT NULL,
    MatchID INT NOT NULL,
    PRIMARY KEY (ClubID, MatchID),
    FOREIGN KEY (ClubID) REFERENCES Clubs(ClubID),
    FOREIGN KEY (MatchID) REFERENCES Matches(MatchID)
)
```

Example:

ClubID	MatchID
1	1
1	5
1	7
2	2
2	5

IV. ENTITY RELATIONSHIP DIAGRAM (ERD)

A. ORGANIZER



This is the Organizer entity, the main controller of the football management system.

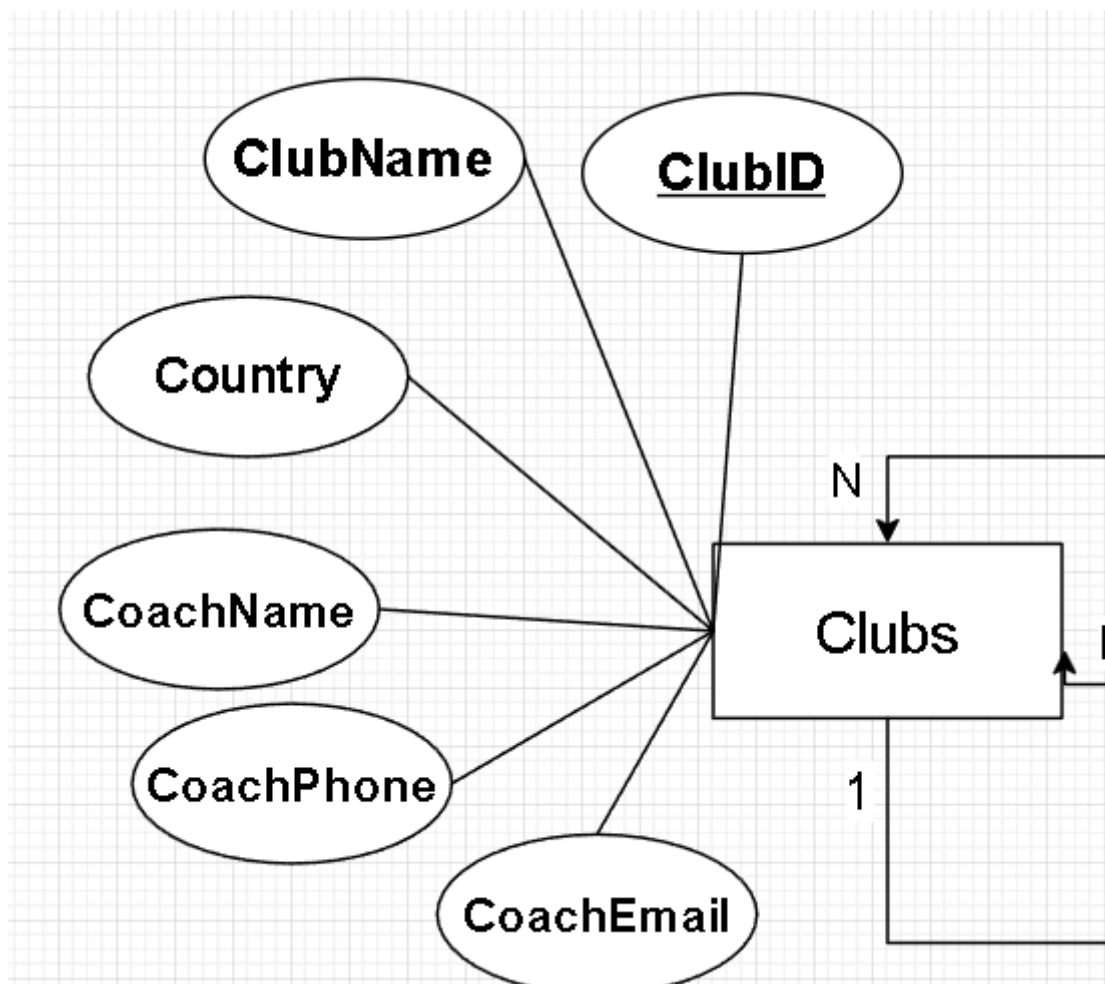
It represents the organization responsible for managing all football clubs, referees, and matches (for example, UEFA).

Organizer	
	OrganizerID
	OrganizerName
	Contact
	Phone
	Email

The attribute OrganizerID is the primary key.

Each organizer has a name (OrganizerName), a contact department/person, and contact information including Phone and Email.

B. CLUBS



The Clubs entity represents all football clubs participating in the tournament.

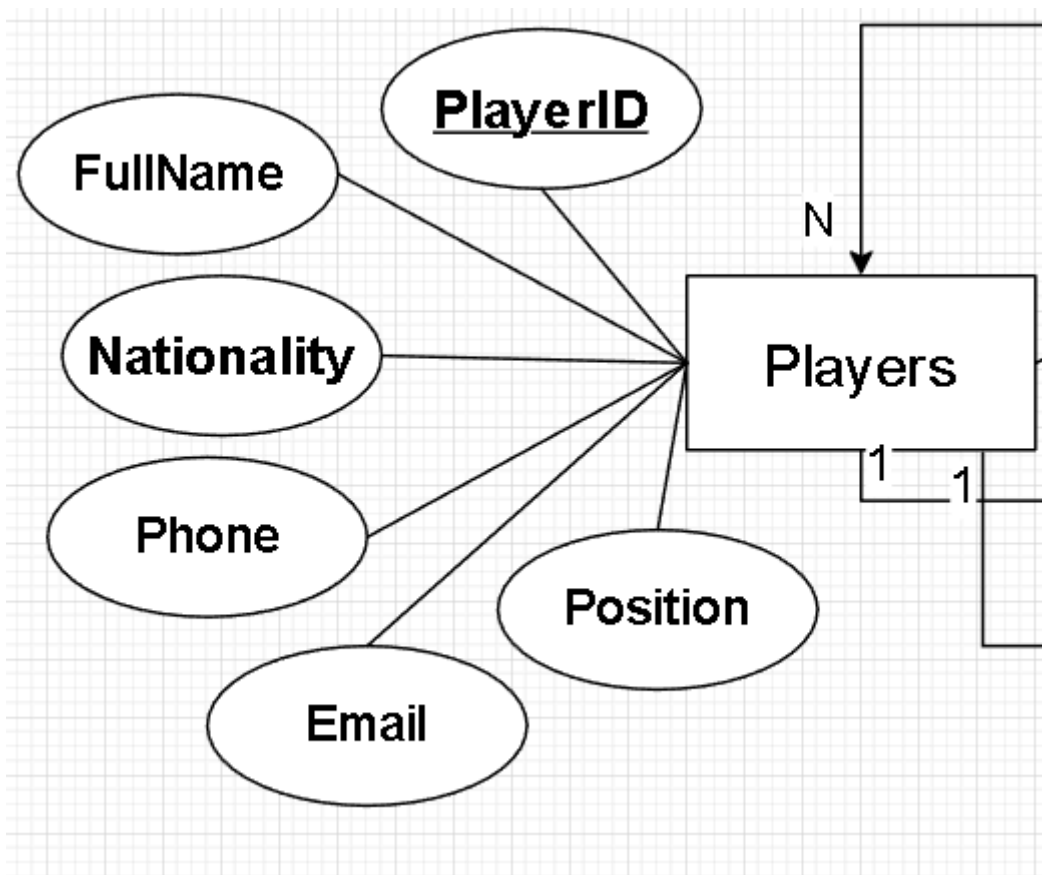
Each club belongs to one organizer and contains details about its coach, contact information, and nationality.

The ClubID is the primary key.

Each club has a name, country, and information about the head coach including name, phone, and email.

Clubs	
	ClubID
	ClubName
	Country
	CoachName
	CoachPhone
	CoachEmail
	OrganizerID

C. PLAYERS



The Players entity stores information about all football players.

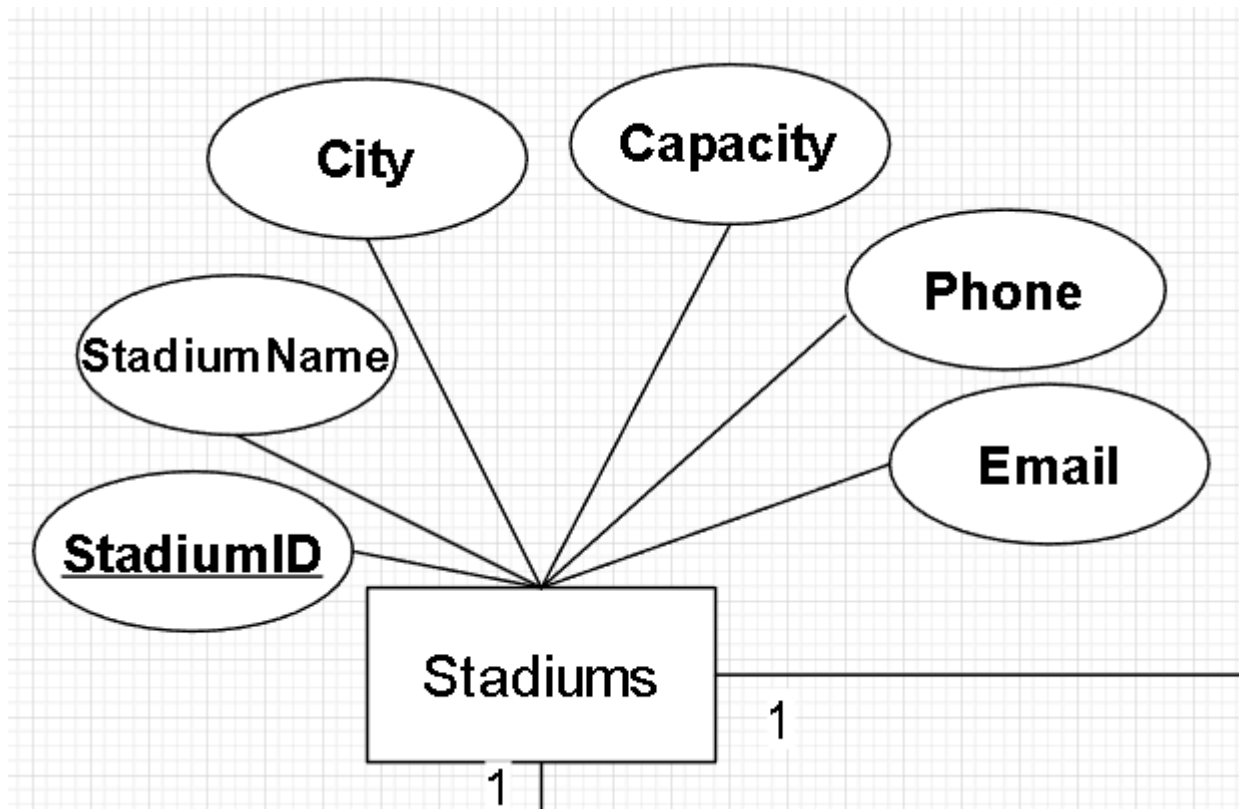
Each player belongs to one club and has personal information such as name, nationality, position, phone, and email.

PlayerID is the primary key.

Each player wears a unique ShirtNumber and plays for one Club.

Players	
	PlayerID
	FullName
	Nationality
	Position
	ShirtNumber
	Phone
	Email
	ClubID

D. STADIUMS

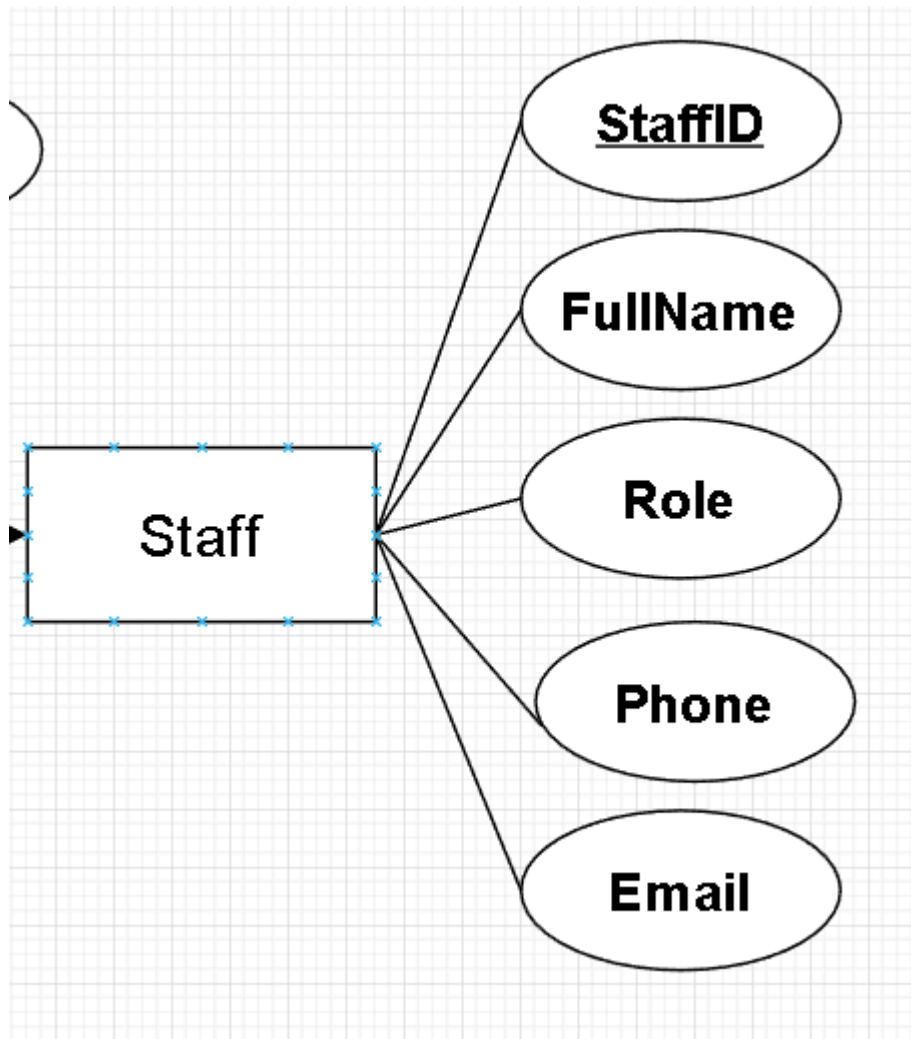


The Stadiums entity represents football stadiums where matches are hosted.

Each stadium has a unique StadiumID, a name, city, capacity, and contact information.

Stadiums	
	StadiumID
	StadiumName
	City
	Capacity
	Phone
	Email

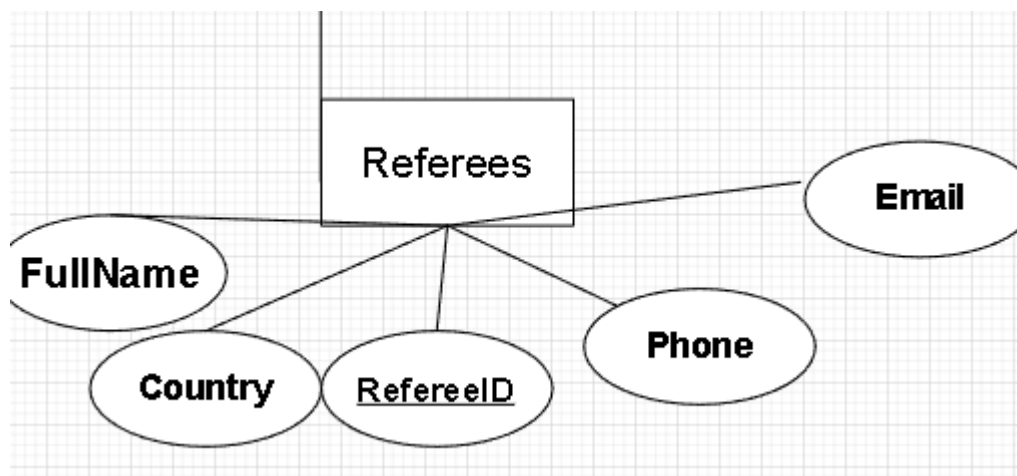
E. STAFF



The **Staff** entity stores information about the employees working in each stadium. Each staff member is assigned to one stadium and has their own **role**, **phone**, and **email**.

Staff	
	StaffID
	FullName
	Staff role
	Phone
	Email
	StadiumID

F. REFEREES

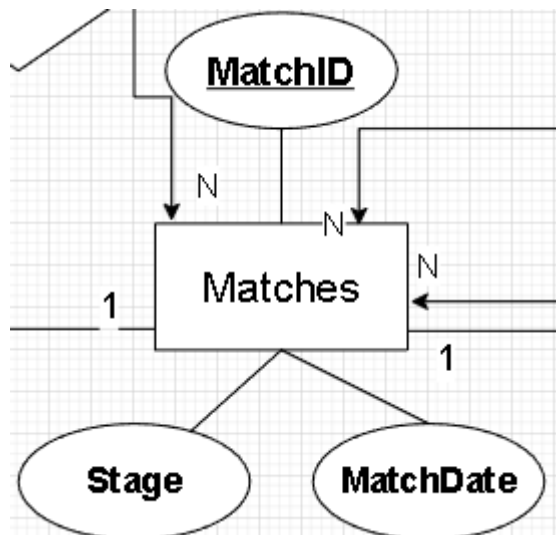


The **Referees** entity stores details about referees who officiate the matches.

Each referee has a name, nationality, phone, and email.

Referees	
	RefereeID
	FullName
	Country
	Phone
	Email

G. MATCHES



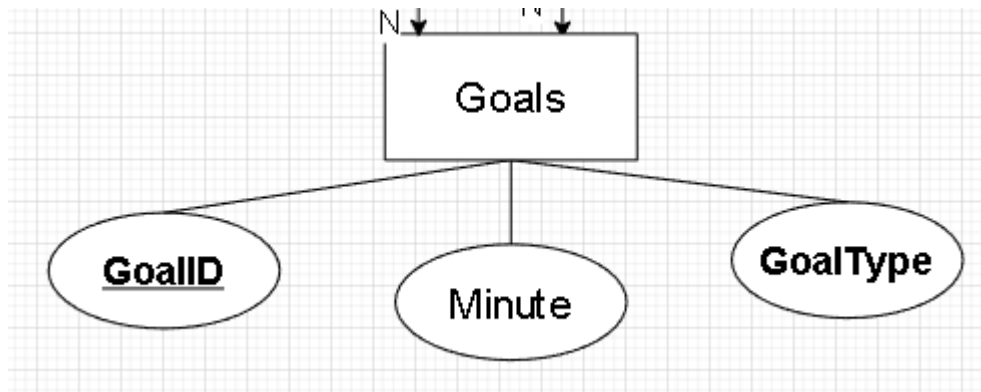
The **Matches** entity represents each football match in the tournament.

It links clubs, stadiums, and referees.

Each match has details such as date, stage and final score.

Matches	
	MatchID
	MatchDate
	Stage
	StadiumID
	RefereeID
	HomeClubID
	AwayClubID
	Score

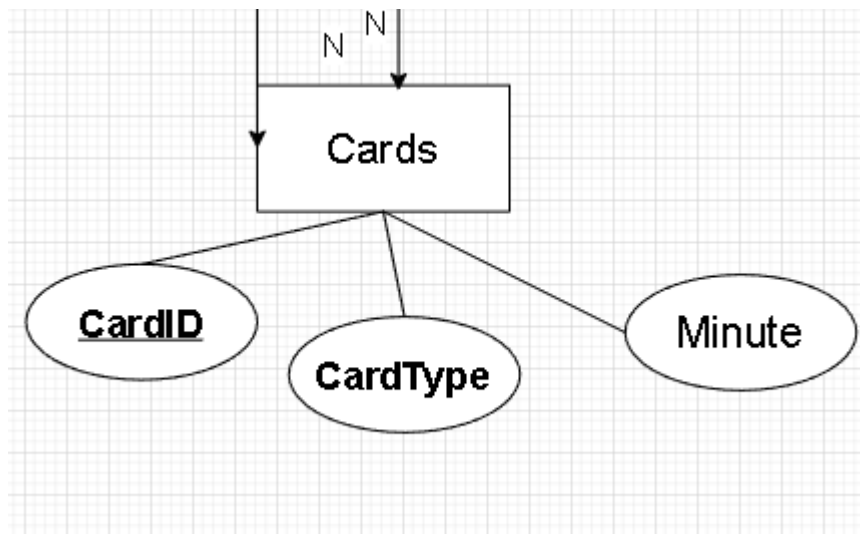
H. GOALS



The **Goals** entity stores information about goals scored during matches. Each goal is linked to both the **Player** and the **Match**.

Goals	
	GoalID
	MatchID
	PlayerID
	Minute
	GoalType

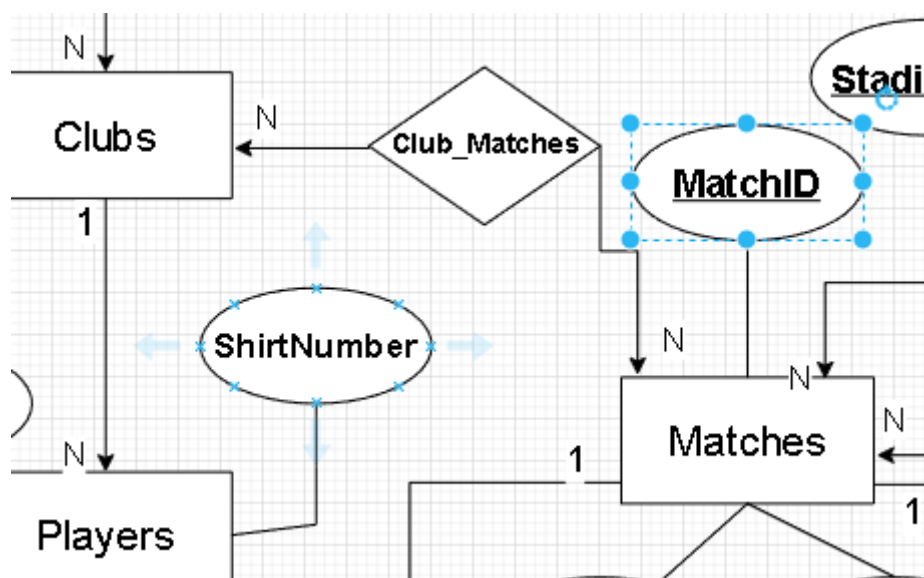
I. CARDS



Each card is linked to a specific **player** and **match**.

Cards	
	CardID
	MatchID
	PlayerID
	CardType
	Minute

J. CLUB_MATCHES

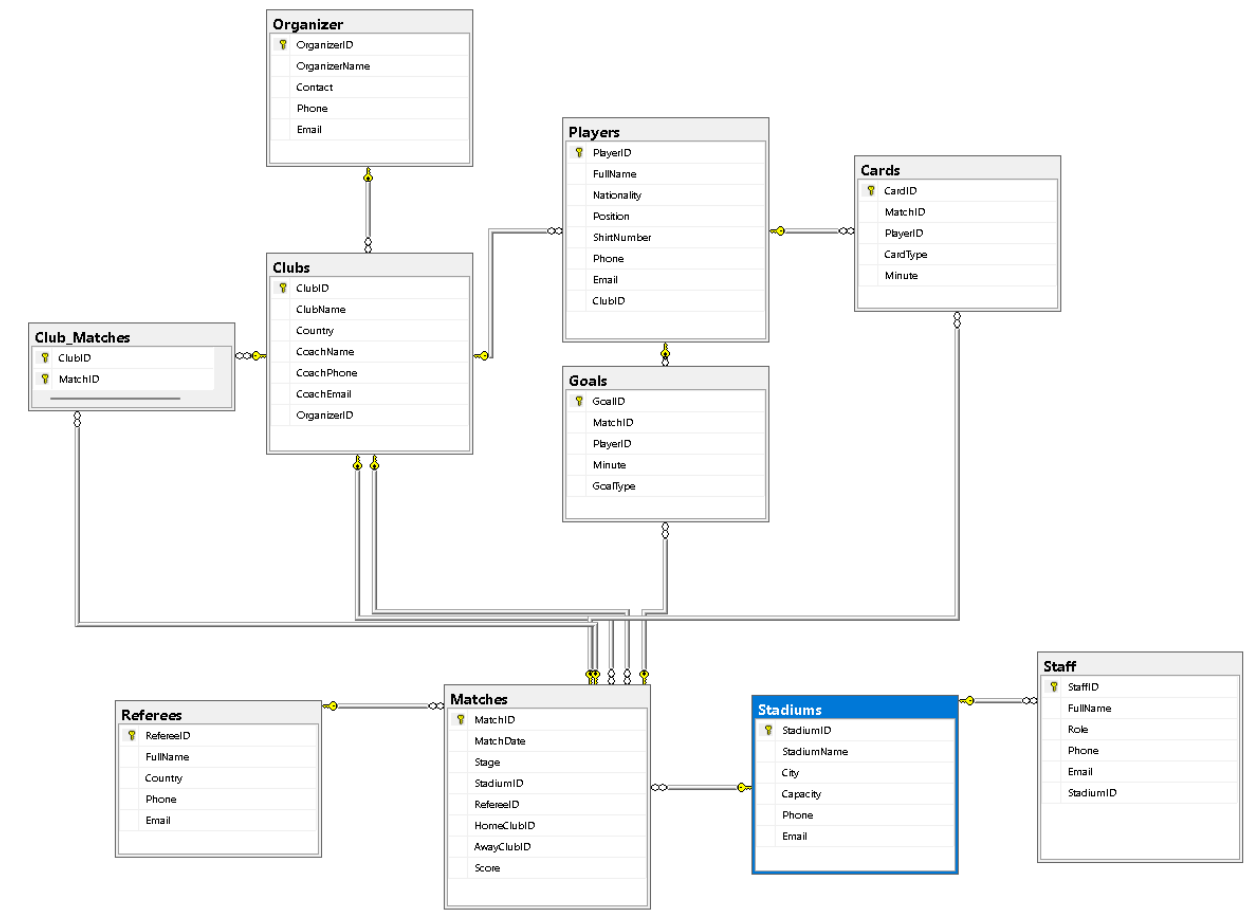


The **Club_Matches** entity is an **association table (many-to-many relationship)** connecting clubs and matches.

It helps identify which clubs participated in which matches.

Club_Matches	
ClubID	
MatchID	

K. FULL DIAGRAM



V. SQL COMMAND

Using Microsoft SQL Server 2016, this server builds intelligent, mission-critical applications using a scalable, hybrid database platform that has everything built in—from in-memory performance and advanced security to in-database analytics.

A. query using order by

Code:

```
SELECT ClubName, Country, CoachName  
  
FROM Clubs  
  
ORDER BY Country ASC, ClubName ASC
```

Result:

	ClubName	Country	CoachName
1	Arsenal	England	Mikel Arteta
2	Liverpool	England	Jurgen Klopp
3	Manchester City	England	Pep Guardiola
4	Paris Saint-Germain	France	Luis Enrique
5	Bayern Munich	Germany	Thomas Tuchel
6	Inter Milan	Italy	Simone Inzaghi
7	Barcelona	Spain	Xavi Hernandez
8	Real Madrid	Spain	Carlo Ancelotti

This query retrieves the list of football clubs and sorts them alphabetically by Country and then by ClubName.

It helps organizers quickly see clubs grouped by their nations.

B. query using inner join

Code:

```
SELECT m.MatchID, c1.ClubName AS HomeTeam, c2.ClubName AS  
AwayTeam, s.StadiumName, m.Score  
  
FROM Matches m
```

JOIN Clubs c1 ON m.HomeClubID = c1.ClubID

JOIN Clubs c2 ON m.AwayClubID = c2.ClubID

JOIN Stadiums s ON m.StadiumID = s.StadiumID

Result:

This query displays all matches with their home team, away team, stadium, and score. The INNER JOIN connects multiple tables to provide a full match overview.

	MatchID	HomeTeam	AwayTeam	StadiumName	Score
1	1	Real Madrid	Liverpool	Santiago Bernabeu	3-1
2	2	Manchester City	Inter Milan	Etihad Stadium	2-0
3	3	Bayern Munich	Barcelona	Allianz Arena	1-1
4	4	Paris Saint-Germain	Arsenal	Parc des Princes	2-2
5	5	Real Madrid	Manchester City	Santiago Bernabeu	2-1
6	6	Bayern Munich	Paris Saint-Germain	Allianz Arena	1-2
7	7	Real Madrid	Paris Saint-Germain	Allianz Arena	2-0

C. query using aggregate functions

Code:

```
SELECT COUNT(PlayerID) AS TotalPlayers  
FROM Players
```

Result:

	TotalPlayers
1	88

This query counts the total number of players recorded in the system.

The **COUNT()** aggregate function is used to calculate the total number of player records in the **Players** table.

Code:

```
SELECT c.ClubName, COUNT(g.GoalID) AS TotalGoals
FROM Goals g
JOIN Players p ON g.PlayerID = p.PlayerID
JOIN Clubs c ON p.ClubID = c.ClubID
GROUP BY c.ClubName
ORDER BY TotalGoals DESC
```

Result:

	ClubName	TotalGoals
1	Real Madrid	7

This query counts how many goals each club has scored.

The **COUNT()** aggregate function and **GROUP BY** clause are used to summarize team performance.

D. query using the group by and having clauses

Code:

```
SELECT s.StadiumName, COUNT(m.MatchID) AS TotalMatches
FROM Matches m
JOIN Stadiums s ON m.StadiumID = s.StadiumID
```

```

GROUP BY s.StadiumName
HAVING COUNT(m.MatchID) >= 2
ORDER BY TotalMatches DESC

```

Result:

	StadiumName	TotalMatches
1	Allianz Arena	3
2	Santiago Bernabeu	2

This query lists stadiums that hosted two or more matches, using the HAVING clause to filter groups based on aggregate values.

E. query that uses a sub-query as a relation

Code:

```

SELECT c.ClubName, g.TotalGoals
FROM (
    SELECT p.ClubID, COUNT(go.GoalID) AS TotalGoals
    FROM Goals go
    JOIN Players p ON go.PlayerID = p.PlayerID
    GROUP BY p.ClubID
) AS g
JOIN Clubs c ON g.ClubID = c.ClubID
WHERE g.TotalGoals = (
    SELECT MAX(TotalGoals)
    FROM (
        SELECT p.ClubID, COUNT(go.GoalID) AS TotalGoals
        FROM Goals go
        JOIN Players p ON go.PlayerID = p.PlayerID

```

```

        GROUP BY p.ClubID
    ) AS MaxGoals
)

```

Result:

Results		Messages
	ClubName	TotalGoals
1	Real Madrid	7

This query finds the club(s) that scored the most goals in the tournament using a subquery as a virtual table.

F. query that uses partial matching in the where clause

Code:

```

SELECT FullName, Nationality, Position
FROM Players
WHERE FullName LIKE '%son%'

```

Result:

	FullName	Nationality	Position
1	Ederson Moraes	Brazil	Goalkeeper
2	Alisson Becker	Brazil	Goalkeeper
3	Andrew Robertson	Scotland	Defender

This query retrieves all players whose names contain the word “son”, using the LIKE operator for partial matching.

G. query that uses a self – join

Code:

```
SELECT  
  
    m1.MatchID AS Match1,  
  
    m2.MatchID AS Match2,  
  
    s.StadiumName  
  
FROM Matches m1  
  
    JOIN Matches m2 ON m1.StadiumID = m2.StadiumID  
  
    AND m1.MatchID < m2.MatchID  
  
    JOIN Stadiums s ON m1.StadiumID = s.StadiumID  
  
ORDER BY s.StadiumName
```

Result:

This query uses a self-join on the Matches table to find pairs of matches that took place in the same stadium.

	Match1	Match2	StadiumName
1	3	6	Allianz Arena
2	3	7	Allianz Arena
3	6	7	Allianz Arena
4	1	5	Santiago Bernabeu
5	1	8	Santiago Bernabeu
6	5	8	Santiago Bernabeu

H. store procedure

Code:

```
CREATE PROCEDURE Show_Match_Info  
    @ClubName NVARCHAR(100)  
AS  
BEGIN
```



```

SELECT m.MatchDate, s.StadiumName, r.FullName AS Referee, m.Score
FROM Matches m
JOIN Referees r ON m.RefereeID = r.RefereeID
JOIN Stadiums s ON m.StadiumID = s.StadiumID
JOIN Clubs c ON (m.HomeClubID = c.ClubID OR m.AwayClubID = c.ClubID)
WHERE c.ClubName = @ClubName;
END

```

EXEC Show_Match_Info @ClubName = 'Real Madrid'

Result:

	MatchDate	StadiumName	Referee	Score
1	2025-04-08	Santiago Bernabeu	Daniele Orsato	3-1
2	2025-04-29	Santiago Bernabeu	Szymon Marciniak	2-1
3	2025-05-28	Allianz Arena	Michael Oliver	2-0

This stored procedure displays all matches played by a specific club, including date, stadium, referee, and score.

I. trigger

Code:

```

CREATE TRIGGER Prevent_Same_Team_Match
ON Matches
AFTER INSERT
AS
BEGIN
    IF EXISTS (

```

```

SELECT 1 FROM inserted
WHERE HomeClubID = AwayClubID
)
BEGIN
    PRINT 'Invalid match: Home and Away teams cannot be the same!';
    ROLLBACK TRANSACTION;
END
END

```

This trigger prevents entering a match where the home and away teams are the same, ensuring logical integrity of data.

WORKS

```

Luu Trung Kien ..... | HE181019 (30%)
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```

THE END

